



Integrating GIS into Sustainable Tourism Strategies

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Received: 25 September 2025 / Accepted: 24 November 2025 / Published: 25 December 2025

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Doi: 10.56345/ijrdv12n3s123

Abstract

Geographic Information Systems (GIS) have become a powerful tool for tourism planning and management, enabling spatial analysis and advanced data visualization. The use of GIS in the tourism sector includes mapping of tourist attractions, accessibility analysis, planning tourist routes and environmental impact assessment. GIS technology helps in the development of sustainable tourism by identifying areas with high tourism potential and minimizing negative impact on the environment through geospatial analysis. This study examines the role of GIS in promoting tourist destinations, analyzing visitor behavior and optimizing tourist services. Furthermore, the integration of GIS with other technologies such as Artificial Intelligence (AI) and Big Data enhances the analysis capabilities, helping to predict tourist flows and improve their experience. The study highlights the importance of using GIS as a strategic tool for developing effective tourism management policies and for promoting a more sustainable and intelligent approach to the development of this sector.

Keywords: GIS, sustainable tourism, spatial analysis, tourism planning, geospatial technology

1. Introduction

In recent years, spatial analysis has become a critical tool for understanding regional disparities in tourism flows. Tourism, as a dynamic sector, requires geographically anchored assessments to support strategic development and infrastructure planning. The integration of GIS (Geographic Information Systems) into tourism studies enables policymakers and researchers to visualize, quantify, and interpret visitor behaviors based on spatial zones, density, and environmental characteristics (Longley et al., 2010).

Tourism is a key economic sector for Albania, showing continuous growth in both domestic and international markets. Understanding the spatial distribution of visitors, especially the distinction between coastal and non-coastal areas, is critical for sustainable tourism planning and infrastructure development. In recent years, attention has increasingly shifted toward analyzing not just visitor numbers, but also their geographic concentration and seasonal flow.

This study focuses on the regional distribution of visitors across Albania for March 2025, using administrative boundary dissolutions into three macro-regions: North, Center, and South. In particular, the research differentiates visitor flow between coastal and non-coastal areas by spatially identifying and isolating coastal administrative units. The central aim is to understand how visitor concentration reflects spatial preferences and accessibility, thus informing future tourism management policies.

Previous research (Hysenaj, 2015) emphasized the value of spatial databases and territorial indexing through digital mapping, particularly in Albania's context. By extending that approach, this study adopts spatial processing tools in

QGIS to calculate, aggregate, and visualize visitor patterns with a focus on percentage distribution and geographical footprint.

This study explores the geographic structure of tourist accommodation in Albania for March 2025 by analyzing official data from the Albanian Institute of Statistics (INSTAT). Emphasis is placed on comparing coastal and non-coastal zones, identifying regional dominance, and generating spatial representations using Geographic Information Systems (GIS). Through this approach, the study provides insights into how tourism load is distributed spatially, supporting future territorial planning and targeted policy interventions.

Geographic Information Systems (GIS) have evolved into essential tools in multiple sectors, including tourism. Their ability to integrate spatial data with decision-making processes enables the identification of high-potential areas, effective service planning, and impact forecasting. This paper focuses on the integration of GIS in sustainable tourism strategies, leveraging geo-spatial tools, web mapping, and predictive modeling.

2. Literature Review

Prior studies (Obe & Hsu, 2010; Longley et al., 2010) emphasize the role of spatial databases (SDBMS) in supporting complex territorial decisions. Hysenaj (2015) has demonstrated how layering and geo-indexing using ArcGIS can support urban planning and business strategy. GIS is now being implemented not only for static mapping but also in dynamic simulations, helping to predict visitor behavior and infrastructure needs.

Zhou and Liu (2022) conducted a bibliometric analysis of GIS-based tourism research, identifying several emerging themes, including the integration of GIS with real-time data systems and machine learning models. These innovations allow tourism planners to predict visitor flows and adapt quickly to environmental pressures or overcrowding issues. Their study also emphasizes the growing use of GIS in resilience planning, particularly in coastal tourism areas prone to environmental degradation.

Another relevant development is the application of Digital Twins in tourism, as discussed by Sampaio de Almeida et al. (2025). These virtual replicas of physical spaces, powered by GIS, allow stakeholders to simulate tourist movements, optimize service locations, and assess the sustainability impact of various scenarios. Integrating Digital Twins with national tourism databases could significantly enhance Albania's ability to forecast tourism patterns and manage growth sustainably.

3. Research Methodology

3.1 Data Source

The data used in this study were obtained from the official INSTAT press release on accommodation establishments for March 2025. The dataset includes the number of visitors, nights spent, and breakdowns by resident and non-resident tourists. The regions analyzed follow the NUTS 3 administrative classification and include 12 counties grouped into three macro-regions: North, Center, and South.

It is important to note that the core analysis in this study is based on data from March 2025, a period outside the peak tourism season. Therefore, the regional ratios between the North, Center, and South - as well as the coastal versus non-coastal dynamics - may vary significantly during high-tourism months such as July and August. Seasonal trends should be considered when interpreting the spatial distribution of visitors.

3.2 GIS Data Processing

Spatial analysis was conducted using QGIS 3.x. The administrative boundary layers were obtained from the GADM database. Coastal zones were defined based on the criteria used by INSTAT - municipalities that fall within 10 km of the coastline.

Key GIS steps included:

- Dissolving units into three macro-regions (North, Center, South). The national territory was classified into three macro-regions (North, Center, and South). Total visitors were aggregated per region, and the percentage of national visitors for each was calculated based on actual counts.
- Selecting coastal municipalities using spatial queries

- Creating buffer zones from the dissolved coastline polygons. Using (municipality level), a buffer and dissolve method was applied to isolate coastal municipalities.
- Overlaying visitor data by zone type (coastal vs. non-coastal)
- Labeling each region with its percentage share of total visitor numbers
- Generating a choropleth map to visualize tourism concentration across macro-regions. The regional distribution of total visitors was represented using a graduated color ramp (light to dark green) to reflect the percentage share per region. Labels were added directly onto the map, indicating regional names and respective percentages (e.g., "Center: 61.3%").

3.3 Analytical Procedure

The visitor data were categorized into coastal and non-coastal zones by manually linking the INSTAT tables with the spatially selected municipalities. Descriptive statistics were calculated for total visitors, and percentages were computed to reflect the share of each zone in relation to the national total. Visual interpretation was strengthened using thematic maps exported directly from QGIS.

Table 1. Regional Distribution of Visitors in Albania

Region	Total Visitors	Percentage (%)
North	25,113	17.9%
Center	86,050	61.3%
South	29,315	20.9%

Data retrieved from INSTAT press releases for March 2025

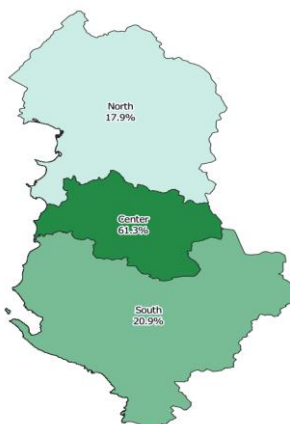


Figure 1. Spatial Distribution of Tourist Visits by Region (March 2025)

Table 2. Regional Visitor Growth Comparison (2024–2025, Period: March)

Region	2024	2025	Growth
North	19,135	25,113	31.24%
Center	71,789	86,050	19.87%
South	24,363	29,315	20.33%

Data retrieved from INSTAT press releases for March 2025

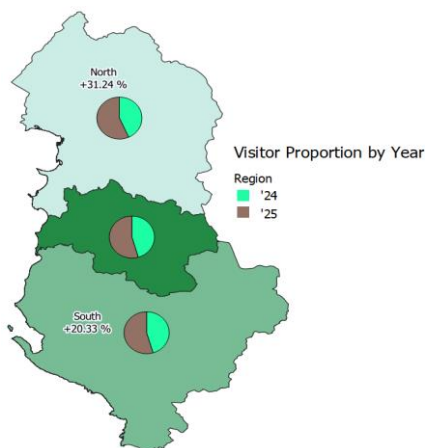


Figure 2. Regional Visitor Proportion and Yearly Growth Comparison (2024–2025)

Table 3. Visitor Distribution by Area Type

Area Type	Total Visitors (August 2024)	Total Visitors (March 2025)
Coastal	491,903	49,701
Non-Coastal	158,690	90,777
Total (Albania)	650,593	140,478

Data retrieved from INSTAT press releases for August 2024 and March 2025

This table underlines that while coastal zones represent a smaller proportion of Albania's landmass, they attract a significant share of non-resident (international) visitors, reinforcing their strategic value. To highlight seasonal disparities, March 2025 data are compared with August 2024 data - the peak tourism month in Albania-based on official INSTAT figures. While these months do not align year-over-year, the aim is to contrast off-season and peak-season spatial dynamics. This comparison provides valuable insights for planning balanced tourism strategies throughout the calendar year.



Figure 3. Distribution of Visitors Between Coastal and Non-Coastal Areas, March 2025

Figure 3's map of visitor distribution between coastal and non-coastal areas highlights the increasing relevance of inland destinations. This spatial insight reveals a potential to rebalance marketing efforts, with zones such as Elbasan or Berat showing moderate but rising visitor trends.

4. Results

The analysis focused on the spatial distribution of tourist activity in Albania for March 2025, using data from INSTAT and GIS mapping. The dataset distinguished between coastal and non-coastal areas, which were extracted and visualized through GIS-based spatial queries and polygon dissolving techniques.

To reflect tourism concentration spatially, administrative units were categorized into three larger regions (North, Center, South), and then further classified into coastal and non-coastal zones based on INSTAT definitions.

The total number of visitors recorded in March 2025 across Albania was: North Region: (17.9%), Center Region: (61.3%), South Region: (20.9%). This confirms a dominant concentration of tourist activity in the Central Region, particularly in and around Tirana and Elbasan.

Using QGIS, Local Administrative Units (LAUs) were identified through spatial selection based on proximity to the shoreline. The results from the official data showed: Coastal areas accounted for 35.4% of all tourist visits. Non-coastal areas accounted for 64.6%, showing a broader distribution of tourism beyond the seaside.

A choropleth map was generated to visualize the regional concentration of visitor percentages. The map labels each region with its share of national tourist activity, offering an immediate spatial interpretation of tourism flow. A second map layer isolates coastal LAUs, clearly illustrating the extent of the tourism belt along Albania's western coast.

This approach enabled both quantitative and geographic analysis, facilitating strategic insights into tourism management by location type.

Furthermore, when comparing the findings from March 2025 with data from August 2024, a significant seasonal disparity becomes evident. In March, coastal areas accounted for 35.4% of all tourist visits, while in August, that figure rose dramatically to 75.6%, confirming the dominant role of seaside tourism during the summer peak season. Non-resident arrivals in coastal areas increased more than tenfold - from 31,522 in March to 349,343 in August. These differences emphasize the seasonal pressure placed on coastal infrastructure and highlight the necessity of GIS tools to monitor, predict, and manage such fluctuations. Incorporating time-series analysis into GIS frameworks can aid in balancing tourism flows between coastal and inland destinations throughout the year.

5. Discussion

The models validate GIS as an enabler for smarter, greener tourism planning. The integration of digital mapping with statistical insights enhances spatial equity, ensures optimized resource use, and supports data-driven policymaking. The web-GIS component allows stakeholders to interact with real-time data, which is vital for responsive planning.

The spatial shift towards non-coastal areas suggests a growing opportunity to diversify Albania's tourism offer beyond its traditional beach destinations. Regions such as Shkodër or Korçë, rich in cultural and natural heritage, could be targeted for investment in eco-tourism or heritage tourism. These shifts also imply a need for improved accessibility, accommodation facilities, and digital promotion of less-known areas.

Notably, the North region experienced the highest year-over-year growth in visitor numbers for March (31.24%), surpassing both the Center and South regions. This trend may reflect a growing demand for highland and nature-based tourism, possibly driven by increased interest in authentic, adventure, and eco-tourism experiences in the mountainous areas of Northern Albania. Destinations such as Theth, Valbona, and Shkodra are becoming increasingly attractive, especially among domestic and regional travelers. This upward trend highlights the strategic importance of developing sustainable tourism infrastructure in the northern highlands, including accommodation, transportation, and digital services, to support and guide this growth responsibly.

In neighboring Croatia, GIS is actively used to monitor coastal over-tourism and reroute tourist flows toward inland destinations. Similarly, Slovenia integrates GIS with mobile data to promote sustainable hiking tourism. Albania could adopt these models to relieve pressure from overvisited coastal zones while promoting underutilized inland destinations.

6. Conclusion

GIS technologies, combined with spatial databases and web-mapping, offer powerful capabilities for sustainable tourism planning. As shown through case models, these tools support strategic decisions by integrating geography with service and infrastructure data. Future work should consider incorporating AI-based prediction models and tourist sentiment analysis for even more adaptive planning.

Moreover, the integration of GIS allows for a multi-scale analysis of tourism impacts - from local infrastructure needs to national-level economic policies. By enabling the visualization of tourism flows, environmental pressure zones, and service distribution, GIS empowers stakeholders to design more equitable and resilient tourism systems.

The Albanian case demonstrates how spatial thinking can uncover imbalances in visitor concentration, highlight untapped regional potential, and guide targeted investments. Ensuring that data accessibility and digital literacy are improved at the local government level will be essential for widespread adoption.

Ultimately, GIS should not be seen only as a technical tool, but as a critical component of a participatory and inclusive tourism strategy that aligns with global goals for sustainable development.

7. Recommendations

Invest in a National GIS Tourism Platform

Encourage the Albanian government and relevant ministries to establish and maintain a centralized GIS platform for tourism management. This platform should integrate real-time data, spatial layers, and analytics to support strategic planning.

Integrate Real-Time and Mobile Tracking Data

Promote the use of mobile tracking technologies (e.g., anonymized phone GPS data) and real-time feedback tools to monitor visitor flows, seasonal peaks, and service usage. This would improve responsiveness and resource allocation.

Support Academic and Vocational Education in Geo-Tourism

Incorporate GIS applications in tourism studies at universities and vocational schools. Develop specialized courses or modules on spatial analysis for tourism planners and managers.

Promote Data Transparency and Open Access

Recommend that publicly funded GIS tourism data (e.g., maps, statistics, geodatabases) be made available through open-access platforms to support academic research, entrepreneurship, and civil society engagement.

Integrate Environmental and Sustainability Indicators

Future GIS-based tourism platforms should include environmental impact layers (e.g., carbon footprint, biodiversity risk zones) to help balance economic growth with environmental protection.

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